TANMOY ROY *PhD*

Kolkata, India • tanmoy.main@gmail.com • (+91) 7003135175, 9830345901 • LinkedIn • Google Scholar

SUMMARY

AI and Machine Learning professional with 19+ years across industry and academia. After 8+ years at SIEMENS, completed a mid-career PhD and post-doctoral fellowship at the University of Johannesburg in speech emotion recognition. Senior Data Scientist at DeliverHealth, leading on-prem small-language-model systems for medical coding, clinical note generation, and medical ASR.

EXPERIENCE

DeliverHealth Solutions

Remote, India [July '24 – Present]

Senior Data Scientist

Lead AI/ML product ideation and organization-wide adoption. Design and deploy on-prem specialist models with product owners and partners including Google and Microsoft. Current bar: models small enough to inspect and honest enough to defer, not merely more fluent.

Key projects: Medical Coding Co-pilot (on-prem SLM fine-tuning; AUROC $\sim$0.60$\rightarrow\sim$0.90 on confidence), Automatic Medical Note Generation (Gemini Pro), RAG-based Medical Correction Engine, Med-ASR (12% medical WER).

Cloudcraftz Solutions

Kolkata, India [Nov '22 – Jun '24]

Senior Data Scientist

Led and mentored a team spanning data science, quantitative analytics, and software engineering, delivering finance-domain systems.

Key projects: Option Trading Strategy Backtest and Analytics, Explainable AI (SHAP/ALE), and Data Visualization Platform.

Veersa Technologies

Kolkata, India [May '22 – Oct '22]

Data Scientist (Consulting)

Primary responsibility was ML model building with both tabular and image data. Other responsibilities involve data engineering, ML model productionizing, and ML team building. Executed projects in the healthcare domain.

Key projects: Insurance Claim Denial Prediction (XGBoost, 89% accuracy), Patient Data De-Duplication, Disease Detection from X-Ray (ResNet34, 84%), Clinical Trial Enrollment Prediction, and Handwritten Medical Form Reader (AWS Textract).

University of Johannesburg

Johannesburg, South Africa [Jul '16 – Sep '21]

Post Doctoral Research Fellow [Oct '19 – Sep '21]

SER deep-learning follow-on, two survey chapters, and research collaborations with King Saud University and BITS Pilani.

Doctoral Research Fellow [Jul '16 – Jun '19]

PhD completed in three years. Thesis on speech emotion recognition: SEGRT feature set and a speech endpoint-detection algorithm.

SIEMENS

KOLKATA, INDIA [Jan '07 – Sep '15]

Associate Consultant [Jan '13 – Sep '15]
Telecom Service Provisioning, CRM integration, and GSM-R Billing systems. Technologies: Java Spring, C/C++, Oracle PL/SQL, Linux.

Sr. System Engineer [Jan '11 – Dec '12]
Electronic Car billing (C/C++, Oracle) and Telecom Call Analysis GUI (Qt/C++).

System Engineer [Jan '09 – Dec '10]
Telecom Billing and Interconnect Charging modules. Technologies: Java, ProC/C++, Oracle.

Associate System Engineer [Jan '07 – Dec '08]
IVR/USSD system support and automated monitoring. Technologies: Java, C++, Linux, shell/awk scripts.

TECHNICAL EXPERTISE

- **Programming Languages:** Python, C/C++, R, Matlab, Java, JavaScript, SQL, Scala, Julia.
- **Libraries:** ollama, langchain, langgraph, unsloth, pytorch, tensorflow, pandas, matplotlib, scipy, opencv, pyspark.
- **Technologies/Platforms:** On-prem GPU inference, Cloud (AWS, GCP), Linux, Docker, MLOps (KubeFlow), REST, SOAP, XML.
- **Databases:** RDBMS (Oracle, PostgreSQL, SQL Server, MySQL), NoSQL (MongoDB, DynamoDB, Cassandra), Cloud DB (Amazon RDS).

EDUCATION

Doctor of Philosophy – University of Johannesburg, 2019. Applied ML & Speech Signal Processing.
Master of Science – Lovely Professional University, 2013. Information Technology.
PG Diploma in Computer Applications – DOEACC Society, 2004.
Bachelor of Science – University of Calcutta, 2002. Economics(Hons), CS, Mathematics.

CERTIFICATES

1. **Deep Learning with PyTorch : Neural Style Transfer**
   Coursera
2. **Neural Network from Scratch in TensorFlow**
   Coursera
3. **Billing and Revenue Management**
   Oracle University
4. **Diploma in Computer Networking**
   CMC Limited

REFERENCES

1. **Lance Mansfield** (LinkedIn)
   Ex CTO and VP DeliverHealth Solutions
   Mob: +1(206)384-2092
   Email: lmansfield17@gmail.com

2. **Nimish Sanghi** (LinkedIn)
   Founder at Cloudcraftz, SOAIS
   Mob: +91 9880569920
   Email: nimish.sanghi@gmail.com

3. **Professor Snehashis Chakraverty** (LinkedIn)
   Professor HAG, Mathematics, NIT Rourkela
   Mob: +919438135400
   Email: sne_chak@yahoo.com , chakravertys@nitrkl.ac.in